

Sri Lanka Institute of Information Technology



IT1040 - Fundamentals of Computing
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Heartrate and Body Temperature Monitoring System.

Progress Report
Group P2-13

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1. Introduction

In our project, The Heartrate and Body Temperature Monitoring System we use several sensors to measure a patients body temperature and heartrate and output them through A LED display. We used Arduino to make this project and the blynk software to get the outputs too other than the LED display mentioned above. With the help of our group members we are currently working on the project rough circuit and we are expecting to develop this circuit to the final prototype.

2. System Architecture

The system is designed to monitor heart rate and body temperature using an **ESP32 microcontroller** as the central unit, which processes data from various sensors and transmits the information to a user interface for real-time monitoring. We are using few specific electronic devices to build this product.

1. **Microcontroller:**

The core of the system is the ESP32 microcontroller, which integrates Bluetooth and Wi-Fi capabilities for wireless data communication. The microcontroller is responsible for reading sensor data and processing it.

Connection: The sensors are connected to the ESP32 through its analog and digital pins for data input. The microcontroller receives data from the sensors, processes it, and either stores it temporarily or transmits it to the user interface via Wi-Fi.

2. **Sensors:**

- **Pulse Sensor (MAX30102):** Used to monitor the heart rate by detecting the pulse rate through a photodetector and two LEDs.

(In the circuit diagram it is shown a different sensor but we have used the MAX30102 sensor in the physical project.)

- **Human Body Temperature Sensor (DS18B20):** Measures body temperature and sends the data to the ESP32 for processing.

(Due to unavailability of this sensor in the shops we decided to continue our project work with the LM35 temperature sensor. We will be replacing the LM35 sensor with the intended sensor in the future when we acquire the sensor from the shops.)

3. **Data Transmission:**

- The ESP32 sends the processed data directly to the mobile/web application via Wi-Fi.
- The system also supports local data display through an OLED screen that shows real-time heart rate and body temperature readings.

4. **Power Supply:**

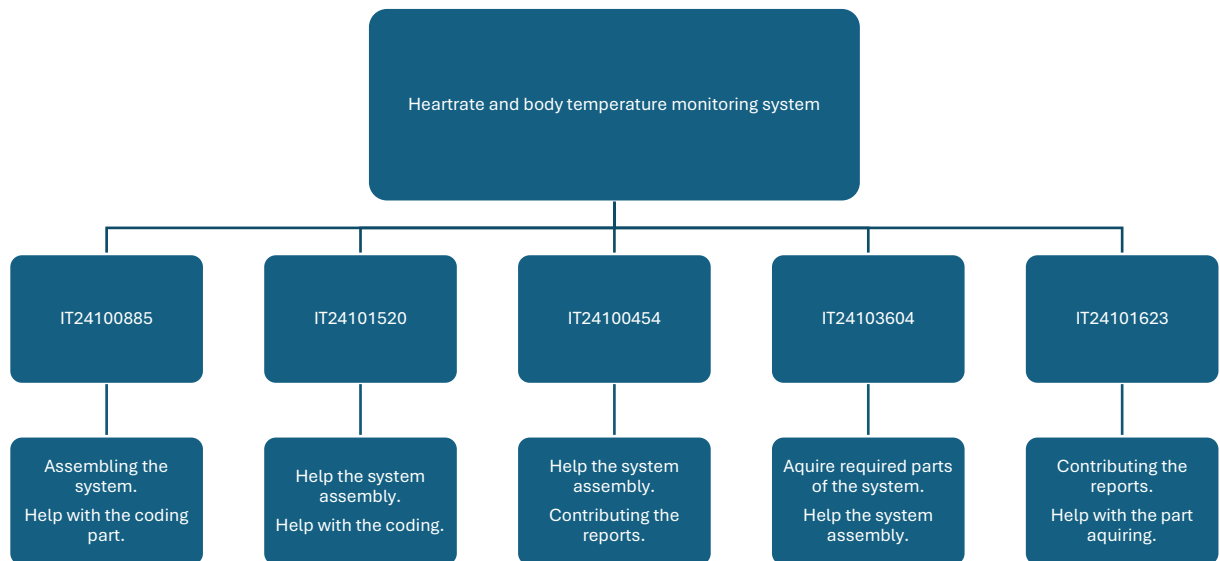
- The system is powered by a battery to ensure portability and continuous operation. A power management system ensures efficient energy use to maintain long-term monitoring.

The system provides continuous monitoring and can trigger alerts for abnormal readings. To do that we are sure this architecture will ensure efficient health monitoring and remote management.

Here is how our system works step by step. It will help to get a clear idea about this product.

1. **Initial Power-Up:** When the system is powered on, the ESP32 boots up, initializing the sensors and establishing a connection to a Wi-Fi network.
2. **Data Collection:** The pulse sensor starts measuring the heart rate, and the temperature sensor starts recording the body temperature. Both sensors send this data to the ESP32.
3. **Data Processing:** The ESP32 processes the raw data, converting the analog signals from the temperature sensor into a temperature reading and reading the heart rate values from the pulse sensor. The ESP32 stores this data temporarily and performs any necessary calculations.
4. **Data Display and Transmission:**
 - **Locally:** The OLED display continuously updates with the current heart rate and temperature readings.
 - **Remotely:** The ESP32 sends the processed data to a mobile or web app through Wi-Fi, where users can see real-time data and get alerts if there are abnormal readings.
5. **Alert Mechanism:** If the system detects abnormal values (e.g., heart rate or temperature outside normal ranges), it triggers an alert within the mobile/web app, prompting the user to take action or seek medical help.

3. Work Breakdown Structure



4. Proof of Work

